

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Dhwan Kumar C / 192511146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dhwan Kumar C

(87)

12

Backtracking search - Technician Assignment.

Problem:

Assign 3 technicians (Tech 1, Tech 2, Tech 3) to 3 machines (Machine X, Machine Y, Machine Z) while satisfying all the given constraints.

Constraints.

1. Tech 1 cannot repair machine Z.
2. Tech 2 must be assigned before Tech 3.
3. Only one technician can be assigned to one machine.
4. Tech 3 cannot repair machine X.
5. Every technician must be assigned exactly one machine.

Initial Domains.

Technician	Possible machines.
Tech 1.	X, Y
Tech 2	X, Y, Z.
Tech 3	Y, Z.

Backtracking steps.

1. Start with an empty assignment.
2. Assign Tech 2 $\rightarrow$ Machine X. (valid).
3. Assign Tech 1 $\rightarrow$ Machine X. (Invalid, already assigned) $\rightarrow$

Backtrack.

4. Assign Tech 1 $\rightarrow$ Machine Y. (valid).
5. Assign Tech 3 $\rightarrow$ Machine Z. (valid).
6. All constraints are satisfied, so the search stops.

Constraint checking.

- * Tech 1 is not assigned to Machine Z.
- * Tech 3 is not assigned to Machine X.
- * Each machine has only one technician.
- * Every technician is assigned exactly one machine.
- * Tech 2 is assigned before Tech 3.

Final Assignment.

Technician	Assigned Machine
Tech 1	Machine Y
Tech 2	Machine X
Tech 3	Machine Z

Conclusion ::

Backtracking systematically checks assignments and backtracks whenever a constraint is violated. The valid schedule obtained is:

Tech 1 $\rightarrow$ Machine Y, Tech 2 $\rightarrow$ Machine X, Tech 3 $\rightarrow$ Machine Z.

2. BFS Search Algorithm in a 5x5 Grid.

Problem ::

A delivery robot must move from Start (S) to Goal (G) in a 5x5 grid while avoiding obstacles.

Constraints ::

1. Allowed movements : up, down, left Right.
2. Cost of each move = 1
3. Robot cannot move through obstacles.
4. Manhattan Distance heuristic :

$$h(n) = |x - x_g| + |y - y_g|.$$

Algorithm:-

Breadth-First Search (BFS) explores nodes level by level using a queue. Since all moves have equal cost, BFS always finds the shortest path.

BFS Steps.

1. Insert the start node (S) into the queue.
2. Remove the first node and expand all valid neighboring cells.
3. Ignore blocked cells and already visited nodes.
4. Add newly discovered nodes to the queue.
5. Repeat until the goal node is reached.

Node Expansion.

Step	Expanded Node	Frontier (Queue)
1.	S	(0,1), (1,0)
2	(0,1)	(1,0), (0,2)
3	(1,0)	(0,2), (2,0)
4	(0,2)	(2,0), (1,2)
5	(1,2)	(2,2)
6	(2,2)	(2,3)
7	(2,3)	(2,4)
8	(2,4)	(3,4)
9	(3,4)	Goal (4,4)

optimal path.

$S \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4) \rightarrow (3,4) \rightarrow G$

Path cost

* Number of moves = 8

* Cost per move = 1.

Total cost =

Advantages of BFS.

* Guarantees the shortest path in an unweighted graph.

* Simple and easy to implement.

* Explores nodes systematically level by level.

Conclusion.

Using the BFS algorithm, the robot successfully reaches the goal by following the shortest valid path with a total cost of 8. Although the Manhattan distance can estimate how close a node is to the goal, standard BFS uses it only as a reference and not for choosing which node to expand next.